

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000827		
	Version:	7.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	24-Apr-2018
Title:	SM DG57-DG517. Cameras replacement.			

DG-57 / DG-517 Service Manual

CAMERAS REPLACEMENT

Procedure description
This procedure describes how replace the reader, samples and reagents cameras. As this cameras are independent, it is divided in different procedures: Procedure 1: Reader camera. Procedure 2: Reagent camera. Procedure 3: Sample camera. Procedure 4: Optics replacement. <i>Note: Document REGD-0000827_v7 replaces document REGD-0000827_v6</i>

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PROCEDURE 1: Reader camera.

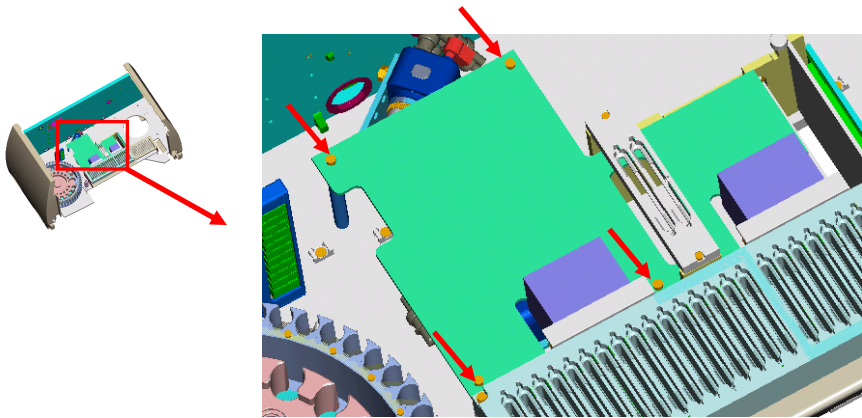
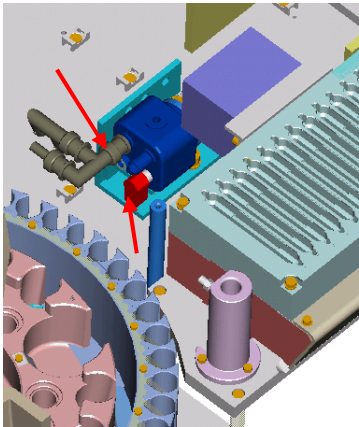
EQUIPMENT

1. A new <i>CAMERA READER</i> spare part.

SET UP

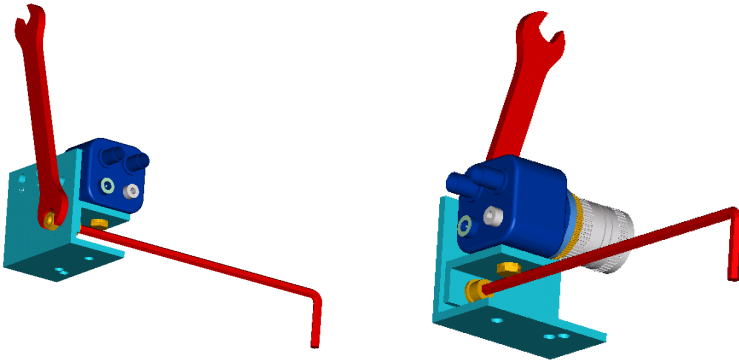
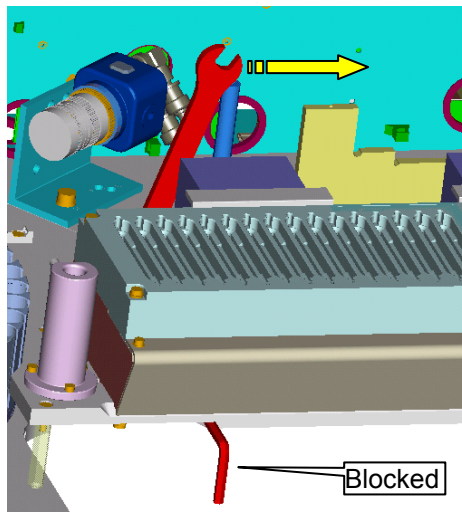
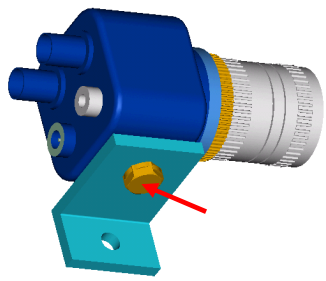
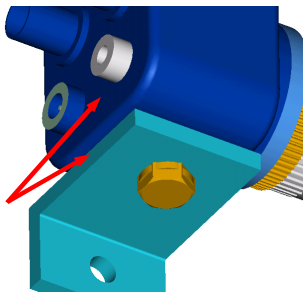
1. Turn off the instrument with the door opened.
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PROCEDURE

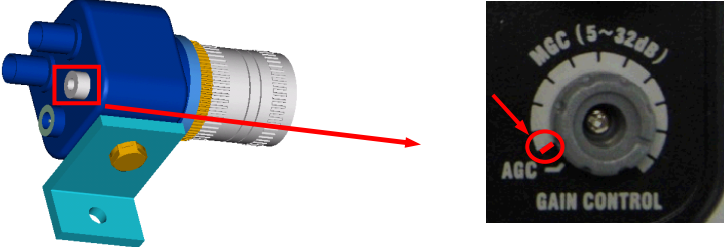
Process	Expected Results
1. Remove the reader cover unscrewing the screws.  Picture 1 2. Disconnect the video and the power connector of the camera.  Picture 2	--- ---

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Process	Expected Results
3. Remove the Waste bottle. Insert the Allen key and the span (as shown in Picture 3).  Picture 3  Picture 4 While the Allen key is blocked manually move the span in the yellow arrow direction (see Picture 4) to unscrew it. 4. Remove the camera from its holder. Place the new camera and screw the nut with a nut driver.  Picture 5  Picture 6 Note the position of the camera, the power switch must be in the lower position (picture 5). The camera must be parallel with the holder (picture 6). Avoid forcing the optics during the screwing operation.	The camera can be removed. ---

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Process	Expected Results
5. Manual Gain Control must be to minimum. Move manually the wheel to the minimum in the Manual Gain Control range. Take care not to overcome "click noise" position.  Picture 7	---
6. Place the camera with its holder in the instrument and fix in it moving the span in the opposite direction of the yellow arrow showed in picture 5. The correct hole to fix for the reader camera is the marked with an "L" and is shown in picture 8.  Picture 8	---
7. Connect the power and video connector (see picture 2). The relative position between the camera connectors and the camera holder is showed in picture 2.	---
8. Apply the REGD-0000812 procedure to adjust the reader camera. Note: During the focus adjustment procedure of the new camera, the optics ring may need to be set to its extreme position in order to set the optimum focus for the acquired image. Should that be the case, an extra washer (included along with the reader camera in spare part 731041 CAMERA READER Ver.3) should be added to the camera assembly. Once the extra washer has been installed, the reader camera adjustment procedure must be performed again.	---
9. Put the reader cover in its place.	---
10. Apply REGD-0000815 procedure to verify the camera's performance.	---

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PROCEDURE 2: Reagents camera.

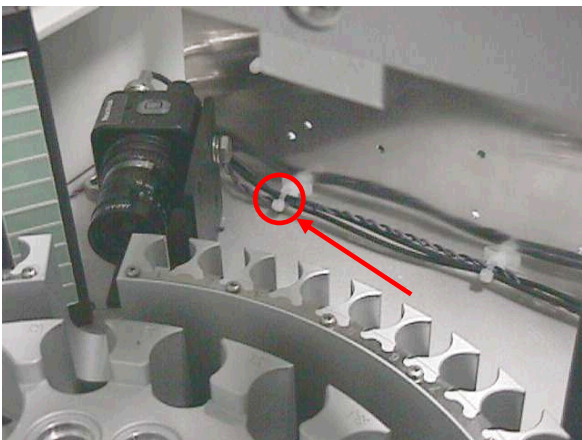
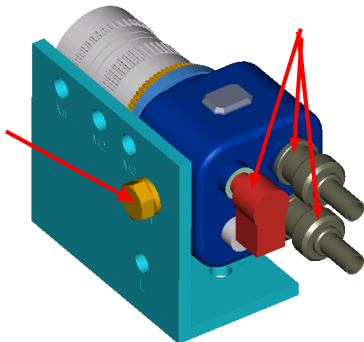
EQUIPMENT

1. A new *CAMERA SAMPLES* spare part.

SET UP

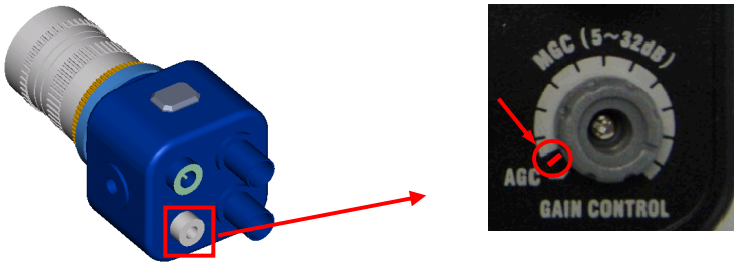
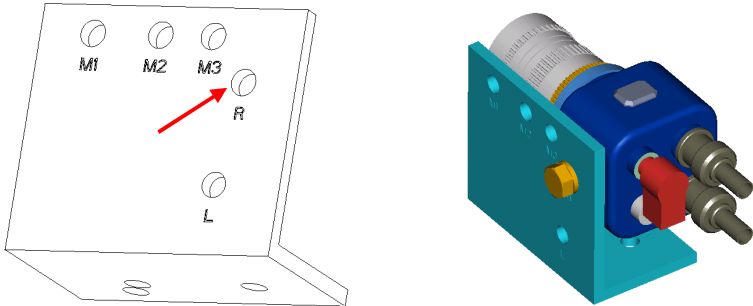
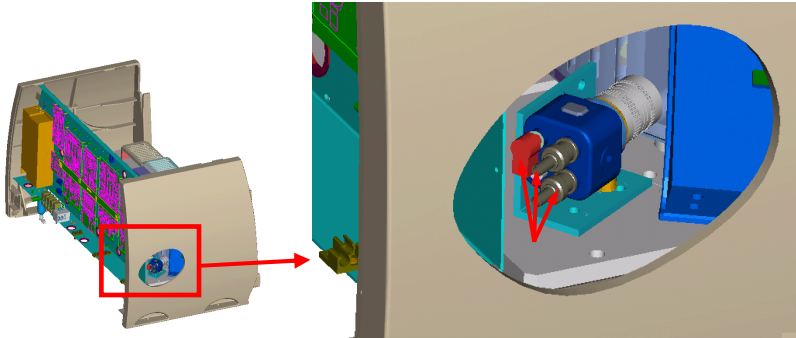
1. Turn off the instrument with the door opened.

PROCEDURE

Process	Expected Results
1. Cut the tie wraps of the camera cables.  Picture 9 2. Remove the camera from its holder unscrewing the bolt.  Picture 10 3. Take the camera and disconnect the video and power connectors. 4. In the new reagent camera the Manual Gain Control must be to minimum. Move manually the wheel to the minimum in the Manual Gain Control range. Take care not to overcome "click noise" position.	--- ---

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Process	Expected Results
 Picture 11 5. Connect the video and power connectors to the new camera and place it in the camera holder to screw the bolt. Be careful to not force the optics in the screwing operation. The correct hole to fix the reagent camera is marked with "R" and is showed in picture 12.  Picture 12 The relative position of the camera connectors with the camera holder is showed in picture 13.  Picture 13 6. Replace the tie wraps showed in the picture 8. 7. Apply the REGD-0000811 procedure to adjust the reagent camera. 8. Apply REGD-0000815 procedure to check the camera.	--- --- --- ---

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PROCEDURE 3: Samples camera.

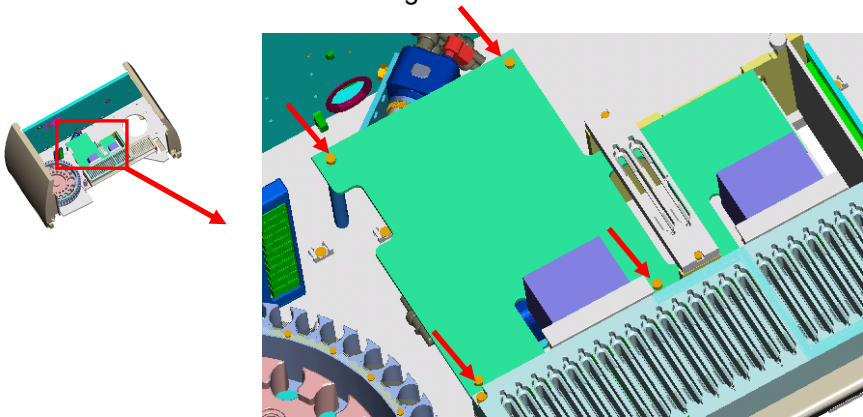
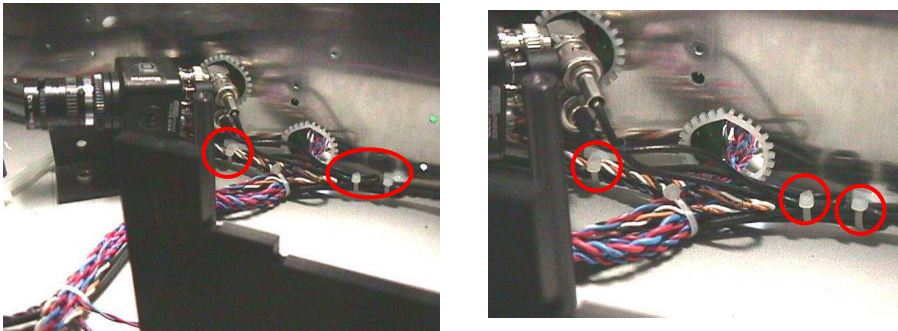
EQUIPMENT

1. A new *CAMERA REAGENTS* spare part.

SET UP

1. Turn off the instrument with the door opened.

PROCEDURE

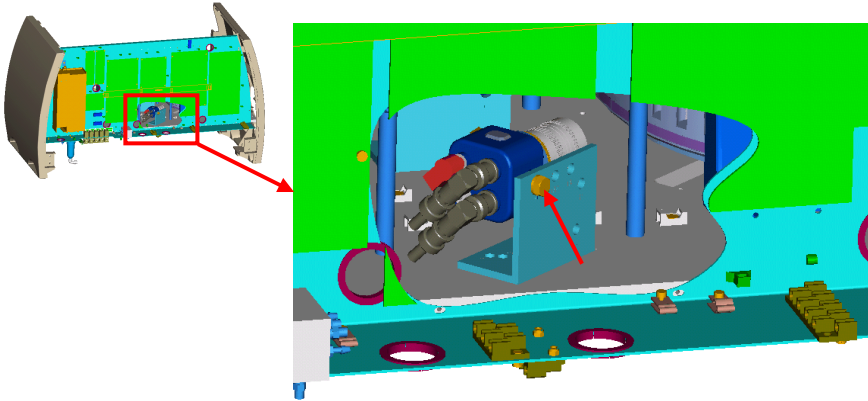
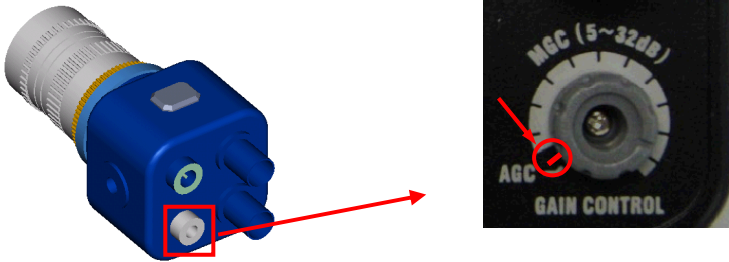
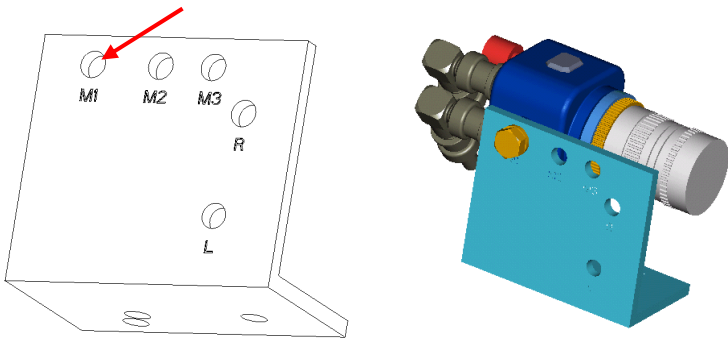
Process	Expected Results
1. Remove the reader cover unscrewing the screws.  Picture 14	---
2. Cut the ties wraps showed in the picture 15.  Picture 15	---
3. Remove the camera from its holder unscrewing the bolt.	---

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Process	Expected Results
 Picture 16 4. Remove the video and power connectors from the camera. 5. In the new samples camera the Manual Gain Control must be to minimum. Move manually the wheel to the minimum in the Manual Gain Control range. Take care not to overcome "click noise" position.  Picture 17 6. Connect the video and power connectors to the new camera and place it in the camera holder to screw the bolt. The correct hole to fix the sample camera is marked with "M1" and is showed in picture 18. Be careful to not force the optics in the screwing operation.  Picture 18 7. Replace the tie wrap showed in the picture 15. 8. Apply the REGD-0000810 procedure to adjust the samples camera. 9. Put the reader cover in its place (see picture 14). 10. Apply REGD-0000815 procedure to check the camera.	--- --- --- --- --- --- --- --- ---

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PROCEDURE 4: Optics replacement.

EQUIPMENT

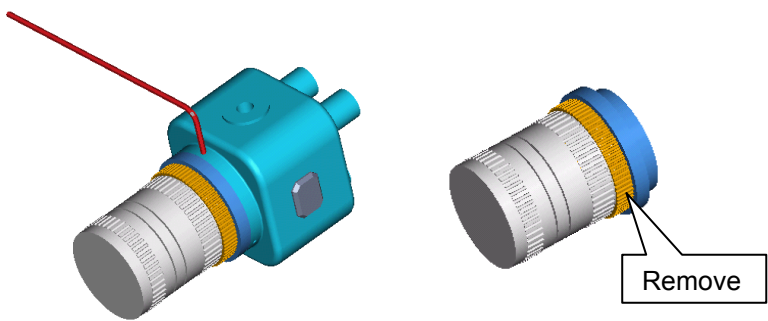
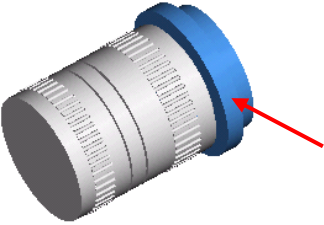
1. An Allen key of 1.27.
2. FOR REAGENTS OR SAMPLES CAMERA:
A new *OPTICS 8mm* spare part.
CAMERA Ver. X (without optics) spare part.
Washers included in *CAMERA OPTICS PLASTIC WASHER* spare part.

FOR THE READER CAMERA:
A new *OPTICS 12mm* spare part.
CAMERA Ver. X (without optics) spare part.
Washers included in *CAMERA OPTICS PLASTIC WASHER* spare part.

SET UP

1. Turn off the instrument with the door opened.
2. Apply the previous procedures (procedure 1, 2 or 3) to remove the camera from the instrument.

PROCEDURE

Process	Expected Results
1. Remove the optics unscrewing the 3 bolts.  Picture 20 From the optics remove the screw adapter (yellow in picture 20). 2. From the new optics remove the adapter unscrewing it.  Picture 21 From the old optics screw manually the screw adapter (yellow in picture 20 and 22) and screw manually the adapter (blue in picture 21 and 22).	--- ---

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Process	Expected Results
Picture 22 3. Insert the washers in the CCD body: Reader camera: 4 washers. An extra washer may need to be added depending on the results of the subsequent adjustment procedure (see Procedure 1 in this document). Samples camera: 4 washers. Reagents camera: 5 washers. Reader Samples Reagent Picture 23 4. For assure a correct placement of the optics place the camera in a plane surface to screw the 3 fixing bolts of the optics. Picture 24 5. Apply the previous procedures (procedure 1, 2 or 3) to replace the camera to the instrument. 6. Adjust the camera. Apply procedures REGD-0000810, REGD-0000811 and REGD-0000812.	--- --- --- ---

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DOCUMENT CHANGE HISTORY

Document change history				
Version	Date	Change	Rationale	
3.0	2005/09/09	- Changed document reference, plus "reagents" for "samples" in procedure 3, step 8. - Added reference to the variable number of washers to be used with reagents and samples cameras' optics; modified reader camera procedure to include visor plate replacement and the reader camera assembly with horizontally adjustable holder.	Document REGD-0000827_v3 replaces document 57054V0M001.	
4.0	2010/09/20	Updated spares part codes of the cameras: - 231817 changed by 232249 (reader) - 231819 changed by 232250 (reagent) - 231818 changed by 232251 (sample) - 231980 changed by 232253 (without optics). Changed the number of washer in the samples camera: - Changed from 3 to 4 washers. - Picture 23. Updated pictures 7, 11 and 17 with the photo of the wheel of the manual gain control range.	The document is updated due to the change of the cameras: new spare parts codes and the change in the number of washers.	
5.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY/MM/DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.	
5.0	2011/04/28	Changed the number of washers for the reagents camera, in step 3 of the optics replacement point.	The number of washer can be 5 or 6 and it is determined by the spare department.	
5.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.	
6.0	July 2011	Document REGD-0000827_v6 replaces document REGD-0000827_v5.	Document reference changed.	
6.0	July 2011	Document title.	Commercial reference substituted by project reference.	
6.0	July 2011	Document header.	Increase information to easy document search.	
7.0	April 2018	Updated description for Procedure 1	Possibility of adding an extra washer for the reader camera.	
7.0	April 2018	Updated description for Procedure 4	Possibility of adding an extra washer for the reader camera.	

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